



Diego F. Vasquez P.

Research Assistant & M.S. in Science in Computer Engineering.
Candidate

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highly motivated researcher currently gaining experience in data analysis, machine learning, and artificial intelligence, with a focus on particle physics and high-energy data transformation. Continuously learning and exploring real-time simulations and the application of deep learning models to complex datasets. Passionate about contributing to innovative research projects and collaborating within multidisciplinary teams. Actively seeking new learning opportunities to acquire additional knowledge and skills.

Experience

Research Assistant *University of Puerto Rico-Mayagüez*
01 2024 - hybrid *Puerto Rico*

- Research Support: Provide comprehensive assistance to principal investigators in data collection, organization, and analysis for key research projects at the CMS particle collider.
- Data Analysis: Conduct data analysis using Python and Jupyter Notebook, extracting vital insights from complex datasets and generating detailed reports for team review.
- Machine Learning: Implement machine learning models for analyzing collision data and improving predictions.
- Data Transformation: Contributed to the NSF project "Innovating AI for Efficient and Insightful Data Transformation" by developing AI-driven techniques for optimizing data workflows.

Conferences and Workshops

PR-LSAMP | ROUSS 2024-2025 *Presenter*
University of Puerto Rico in Aguadilla

- Presented a poster titled "Primary Vertex Detection in CMS Proton-Proton Collision Open Data using Wasserstein Distance"

The 2024 Summer Workshop *Presenter*
Bartos Theater, MIT List Visual Arts Center, Boston August 12 - 16, 2024

- Presented a poster titled "Jet Classification using Dimensionality Reduction and Deep Learning"

Languages

Spanish (native) English (currently learning)

Skills

Python Programming, Data Analysis, Machine Learning, Team Collaboration, Computer and Network Setup, Continuous Learning, Technical Skills, Personal Skills.

Interests

Programming, Research, Artificial Intelligence, Machine Learning.

Education

M.S. in Science in Computer Engineering *University of Puerto Rico Mayagüez*

January 2024 - In progress *Puerto Rico* **Systems Engineer**
Universidad Nacional Abierta y a Distancia UNAD
August 2020 - December 2023 *Colombia*

- Completed a semester with 10 credits in a specialization in Data Science and Analytics as a graduation requirement.
- Participated in the Rally Latinoamericano de Innovación in 2022 and 2023, showcasing innovative solutions to real-world problems.

Courses Completed *Platzi, Cisco, IBM*

- Structured Programming Course (August 2020) Platzi
- Introduction to Computational Thinking with Python Course (August 2020) Platzi
- Fundamentals of Software Engineering (August 2020)
- Artificial Intelligence Workshop with IBM Watson (IBM, November 2021)
- Linux Server Administration Course (December 2021) Platzi
- Introduction to Terminal and Command Line Course (August 2021) Platzi
- Python Essentials 1 (Cisco, June 2023)

Projects

Innovating AI for Efficient and Insightful Data Transformation *Research Assistant*

November 2023 - Present

- Contributed to the development of advanced AI-driven data transformation techniques as part of an NSF-funded project.
- Collaborated with a multidisciplinary team to optimize data workflows and ensure efficient data analysis processes.

Jet Classification Using GNNs *Research Assistant*
February 2024 - Present

- Worked on classifying b, c, and light jets using collision data from the CMS detector, employing GNNs and dimensionality reduction techniques like PCA and Autoencoders. Improved classification accuracy by implementing attention layers in SetToGraph model training.